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Revamp wasi example and related docs #9788

Merged

alexcrichton merged 22 commits into `bytecodealliance:main` from `ifsheldon:revamp-wasi` on Jan 8

Conversation 29

Commits 22

Checks 39

Files changed 12

 ifsheldon commented on Dec 12, 2024 • edited ▾ Contributor

Closes [#9777](#)

~~For now, this is a draft PR because there's a runtime issue in the revamp example. Please see the FIXME in `examples/wasi/main.rs`. There I want to invoke the exported `wasi_cli/run@0.2.0` function, which is the main function in `examples/wasi/wasm/wasi.rs`. Probably [@alexcrichton](#) or [@fitzgen](#) can help a bit? Thanks!~~

~~When the runtime issue is fixed, I will try to revamp more sections in the related doc in following commits~~

Updated:

This PR revamps the examples `wasi` and `wasi-async` with latest updates from `wasmtime` and `wasmtime-wasi` and uses WASIp2 APIs by default. Related documentation is also updated.



 1

  update `examples/wasi/main.rs` and related doc with a runtime issue ✖ e1becf1

 github-actions bot added the `wasmtime:docs` label on Dec 12, 2024

 alexcrichton reviewed on Dec 13, 2024

[View reviewed changes](#)

`docs/examples-rust-wasi.md` Outdated Show resolved

`examples/wasi/main.rs` Outdated Show resolved

-   simplify building instruction c040b6c
-   fix runtime issue ✖ 7098360

ifsheldon commented on Dec 13, 2024

Contributor

Author

But now there comes another issue: this hello-world example is just too specific to `wasi:cli`, since we used a specific `wasmtime_wasi::bindings::sync::Command` to run a `wasi:cli/command`. For a more general component, we need to add another example or at least give a pointer.

BTW, I still don't know why I could not get a `Func` to execute the `run` function in `wasi:cli/run`.



alexcrichton commented on Dec 14, 2024

Member

Personally I think it's reasonable to start with hello-world like this and then graduate up to other examples. For example this page could link to the `bindgen_examples` page and that could get expanded with a custom world for example (IIRC there may already be one there?)

BTW, I still don't know why I could not get a `Func` to execute the `run` function in `wasi:cli/run`.

You needed another invocation of `get_export`. You loaded the exported instance but you'll need to load the exported function of that instance as well.



 ifsheldon added 3 commits [4 months ago](#)

-   add a comment and cargo fmt 8848142
-   revamped wasi-async 5ab2c86
-   remove the example of custom host states and add references ✔ 0df6756

  ifsheldon marked this pull request as ready for review 4 months ago

  ifsheldon requested review from **a team** as code owners 4 months ago

  ifsheldon requested review from **dicej** and removed request for **a team** 4 months ago

OK, I think this draft PR seems good to me now, so I turned it into a formal PR.



🔗  add calling function dynamically

❌ 62703f1

ifsheldon commented on Dec 14, 2024 • edited ▼

Contributor

Author

You needed another invocation of `get_export`. You loaded the exported instance but you'll need to load the exported function of that instance as well.

OK, got that. I also added related code in the example to demonstrate how to do that.

BTW, do you know why [ComponentNamedList](#) is implemented for tuple `(A1,)` but not `A1` ? Should I file another issue for this? This is bit counter-intuitive when I want to write something like `func.typed::<(), Result<(), ()>>(&store)`, which `rustc` complains about not implementing `ComponentNamedList`. I really didn't know how to fix this error until I lookup `Implementations` on `Foreign Types` section of [ComponentNamedList](#).



🔗  cargo fmt

✅ 4c6cce2

🔗  ifsheldon mentioned this pull request on Dec 14, 2024

Add async example to bindgen_examples #9822

🔗 Merged

🔗  add post_return

✅ ceead4a

👁 alexcrichton reviewed on Dec 16, 2024

[View reviewed changes](#)



alexcrichton left a comment

Member

Thanks!

This is bit counter-intuitive when I want to write something like `func.typed::<(), Result<(), ()>>(&store),`

IIRC there were coherence with that trait impl, but I could be misremembering as well.



docs/examples-rust-wasi.md Outdated

Show resolved

docs/examples-rust-wasi.md Outdated

Show resolved

docs/examples-rust-wasi.md Outdated

Show resolved

examples/wasi/main.rs Outdated

Show resolved

ifsheldon added 3 commits [4 months ago](#)

- update wording 953740a
- add expects a936a8a
- change wording from "module" to "component" 4e4304a

alexcrichton approved these changes on Dec 17, 2024

[View reviewed changes](#)

alexcrichton added this pull request to the [merge queue](#) on Dec 17, 2024

github-merge-queue bot removed this pull request from the [merge queue](#) due to failed status checks on Dec 17, 2024

[View details](#)

ifsheldon commented on Dec 17, 2024 • edited

Contributor

Author

Oops. CI complains about file not found `target/wasm32-wasip2/debug/wasi.wasm`, so I guess CMakeLists also needs to be updated.

Furthermore, I dugged into `examples/wasi/main.c` and related `c-api` crate. It seems `c-api` crate also needs some renovation.....

For example, `c-api` exposes this function, which links wasip1 interfaces, but no other functions link wasip2 interfaces.

I think removing `preview1` feature from this line and fixing whatever breaks should be enough. This introduces API breaking changes, though.

[wasmtime/crates/c-api/Cargo.toml](#)

Line 36 in [db4bd21](#)

```
36    wasmtime-wasi = { workspace = true, optional = true, features = ["preview1"] }
```

I'd rather like to skip this error first and open another tracking issue.



  `fix CMakeLists.txt`

✓ c8e6740

8 hidden items

[Load more...](#)

alexcrichton commented on Dec 20, 2024

Member

Ah I better understand what was going on now. Instead of replacing the current example which I think is still useful for `wasm32-wasip1` targets, could this perhaps rename the prior `wasi` example to `wasip1` and add this new example under `wasip2`? There then wouldn't be a C example of `wasip2` because that's not supported yet.



 **ifsheldon** added 3 commits [4 months ago](#)

  `rename wasi examples`

469548c

  `fix wording`

✗ 27043b4

  `fix format`

✓ 6e86762

ifsheldon commented on Dec 20, 2024

Contributor

Author

OK, I've brought back `wasip1` examples.



[view reviewed changes](#)

  **alexcrichon** added this pull request to the [merge queue](#) on Dec 21, 2024

  **github-merge-queue** bot removed this pull request from the [merge queue](#) due to failed status checks on Dec 21, 2024

[View details](#)

ifsheldon commented on Dec 21, 2024 • edited ▾

[Contributor](#)

[Author](#)

This [test failure](#) is weird to me. I have cleaned all caches and run the commands in this test manually on my Mac but I saw no failures.



  Merge branch 'main' into revamp-wasi, upstream sync

✓ 4cc8be9

  **alexcrichon** added this pull request to the [merge queue](#) on Jan 6

  **github-merge-queue** bot removed this pull request from the [merge queue](#) due to failed status checks on Jan 6

[View details](#)

alexcrichon commented on Jan 7

[Member](#)

Have you tried executing only the steps that the [failed CI job](#) is executing?



ifsheldon commented on Jan 7

[Contributor](#)

[Author](#)

Yes. Here is my complete log of my terminal outputs. Search for "\$" to see my commands:

1. cargo clean
2. cmake -Sexamples -Bexamples/build -DBUILD_SHARED_LIBS=OFF copied from CI
3. cmake --build examples/build --config Debug copied from CI
4. cmake -E env CTEST_OUTPUT_ON_FAILURE=1 cmake --build examples/build --config Debug --target test copied from CI

[log.txt](#)



ifsheldon added 2 commits [3 months ago](#)

- 🔗 Merge branch 'main' into revamp-wasi 87f8ab8
- 🔗 add wasm32-wasip2 target ✓ 0aa610f

ifsheldon commented on Jan 7 • edited ▾

Contributor

Author

I think CI silently swallow the errors caused by a target that was not added. My local run is successful because I have added `wasm32-wasip2` target myself before and everything just compiled.

[wasmtime/.github/actions/install-rust/action.yml](#)

Line 67 in [fc7ef8d](#)

```
67      run: rustup target add wasm32-wasip1 wasm32-unknown-unknown
```

This is used by CI, but it does not add `wasm32-wasip2` target. CI should have given better errors stating that a target is not found.



🔗 ifsheldon mentioned this pull request on Jan 7

CI silently swallows errors and tests forget to check return values #9932

🔒 Closed

🕒 alexcrichton added this pull request to the [merge queue](#) on Jan 7

🕒 github-merge-queue bot removed this pull request from the [merge queue](#) due to failed status checks on Jan 7

[View details](#)

alexcrichton commented on Jan 7

Member

Could the wasip2 target be installed for just this one CI job perhaps instead of all of them?



🔗 ifsheldon added 2 commits [3 months ago](#)

- 🔗 Revert "add wasm32-wasip2 target" ... 0beb88a

  alexcrichon added this pull request to the [merge queue](#) on Jan 8

Merged via the queue into [bytecodealliance:main](#) with commit **5030709** on Jan 8
39 checks passed

[View details](#)

  ifsheldon deleted the `revamp-wasi` branch 3 months ago

 chobermaier added a commit to chobermaier/wasmtime that referenced this pull request on Jan 8

 fix example includes for [bytecodealliance#9788](#) 04bc991

 chobermaier mentioned this pull request on Jan 8

fix broken example includes for #9788 #9948

 Merged

 github-merge-queue bot pushed a commit that referenced this pull request on Jan 8

 fix example includes for [#9788](#) ([#9948](#)) ✓ 4ad53fa

Reviewers

 alexcrichon	✓
 dicej	✓

Assignees

No one assigned

Labels

[wasmtime:docs](#)

Projects

None yet

Milestone

No milestone

✓ Renovate the WASI example

3 participants

